

Combinational Circuits :

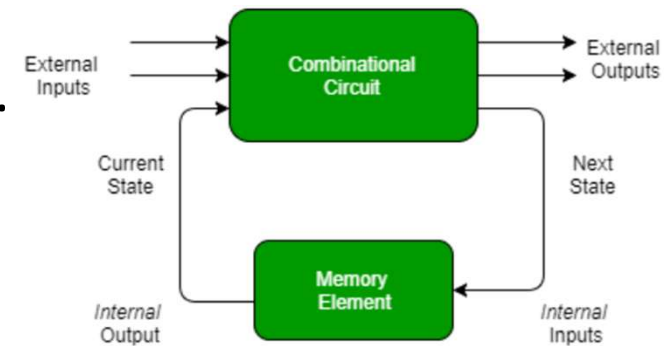
- Time dependant circuits
- The output of the circuit depends only on Present inputs .
- $\text{Output} = f(\text{Present inputs})$
- Speed is fast.
- There is no feedback between input and output.
- Elementary building blocks: Logic gates
- Easy to design
- Combinational circuits don't have capability to store any state
- Examples: Encoder, Decoder, Multiplexer, Demultiplexer



Figure: Combinational Circuits

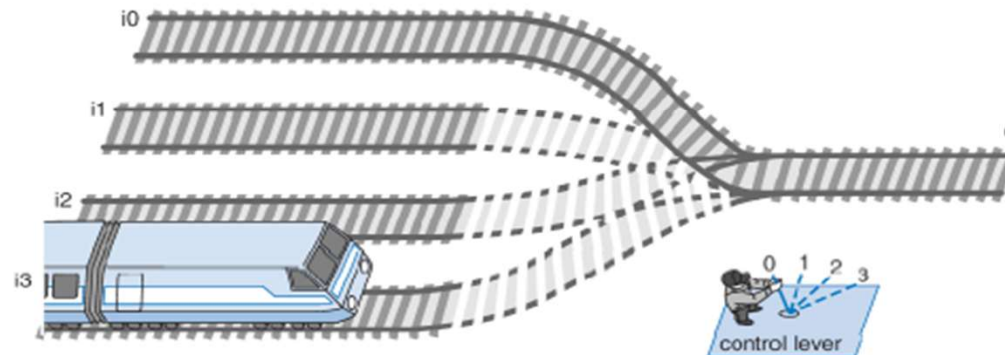
Sequential Circuits :

- The output of the circuit depends on Present inputs and past inputs / outputs of the circuit .
- Sequential circuits have capability to store a state (memory element)
- $\text{Present Output} = f(\text{Present inputs, Past output})$
- There is a feedback between input and output.
- Clock and time dependent circuit
- Elementary building blocks: Flip flop
- Design complicated as compared
- Builds on Combinational logic
- Examples: Flip flops, Counters, Shift registers, Finite state machines

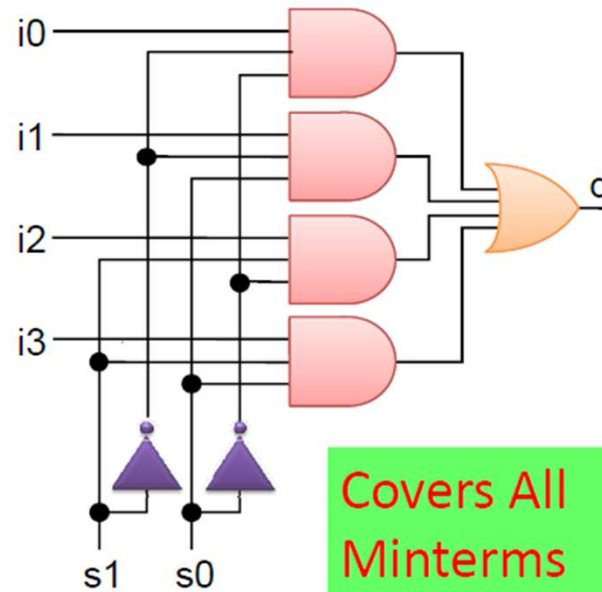
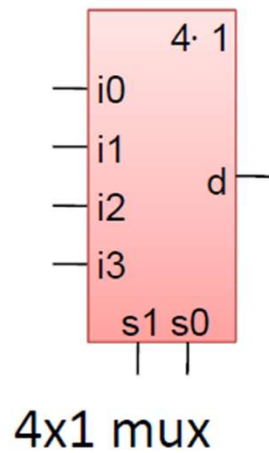


MULTIPLEXERS

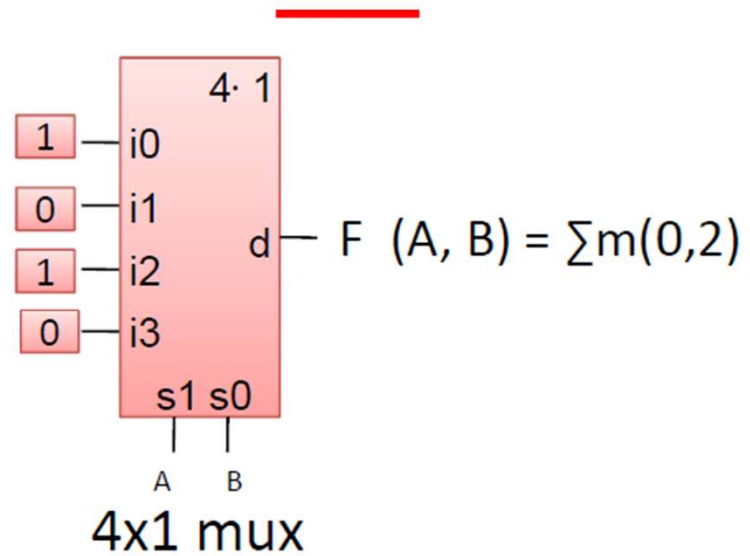
- Mux: Another popular combinational building block
 - Routes one of its N data inputs to its one output, based on binary value of select inputs
 - 4 input mux $\rightarrow$ needs 2 select inputs to indicate which input to route through
 - 8 input mux $\rightarrow$ 3 select inputs
 - N inputs $\rightarrow \log_2(N)$ selects
 - Like a railyard switch



MULTIPLEXERS

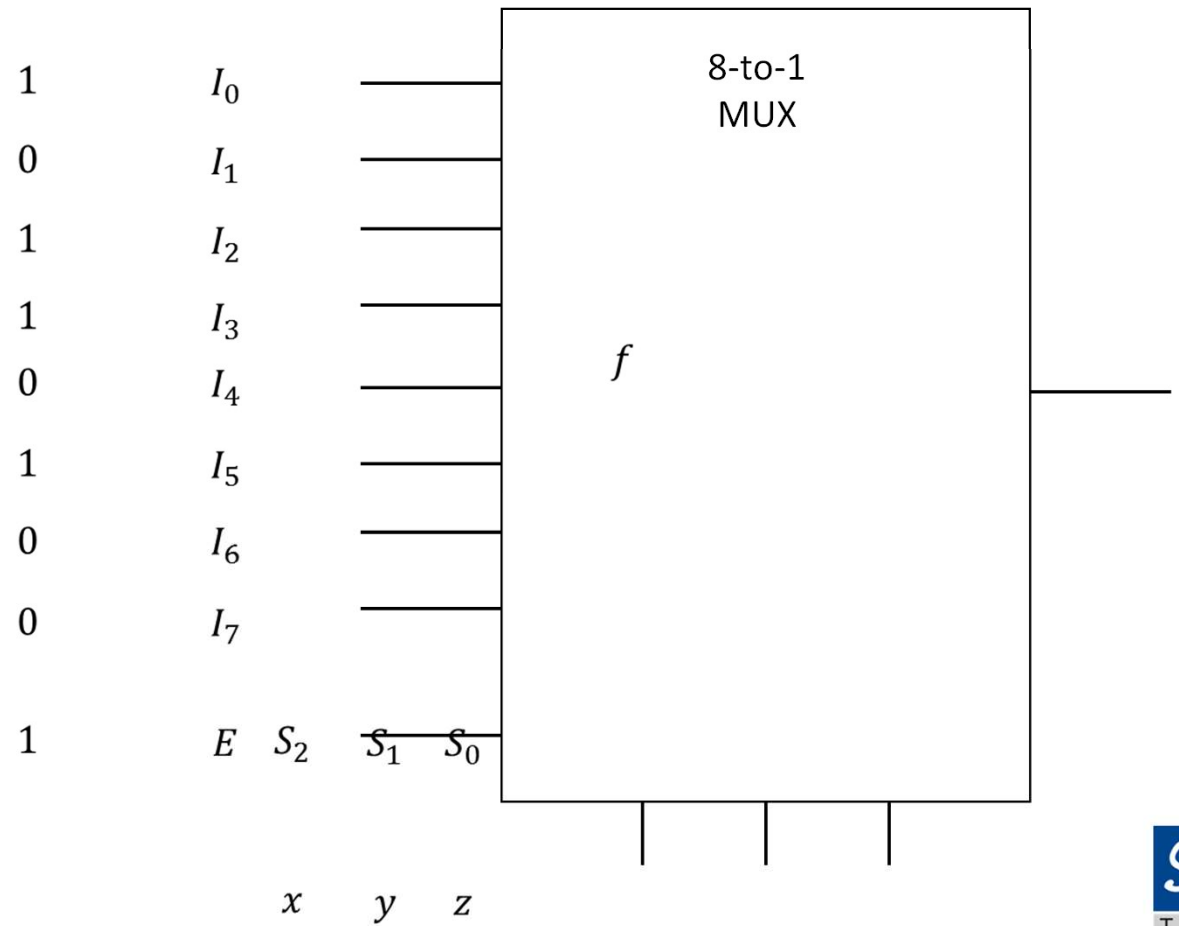


MULTIPLEXERS



Example:

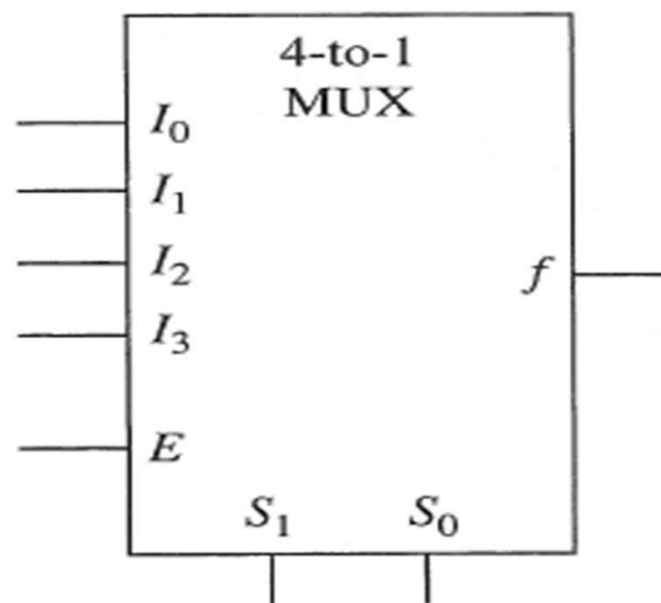
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



Example

0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

$\bar{z}$
1
 z
0
1

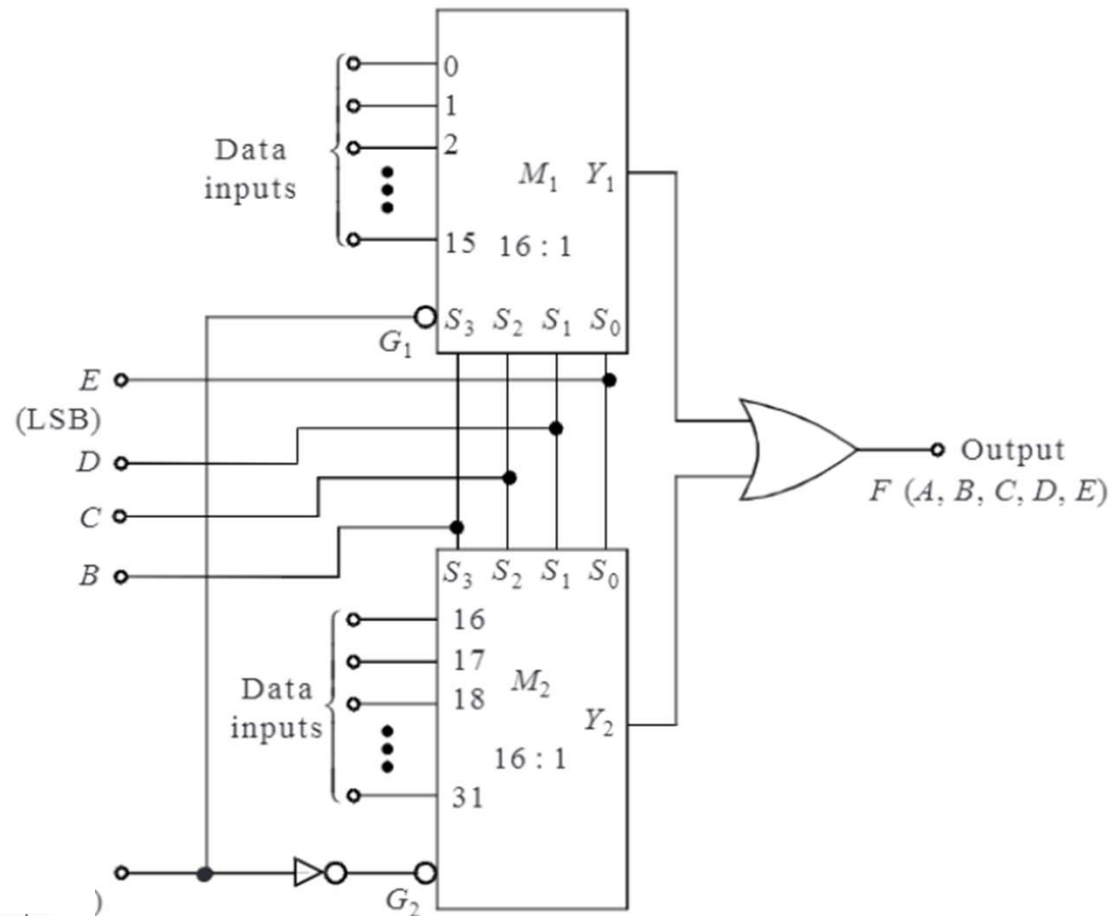


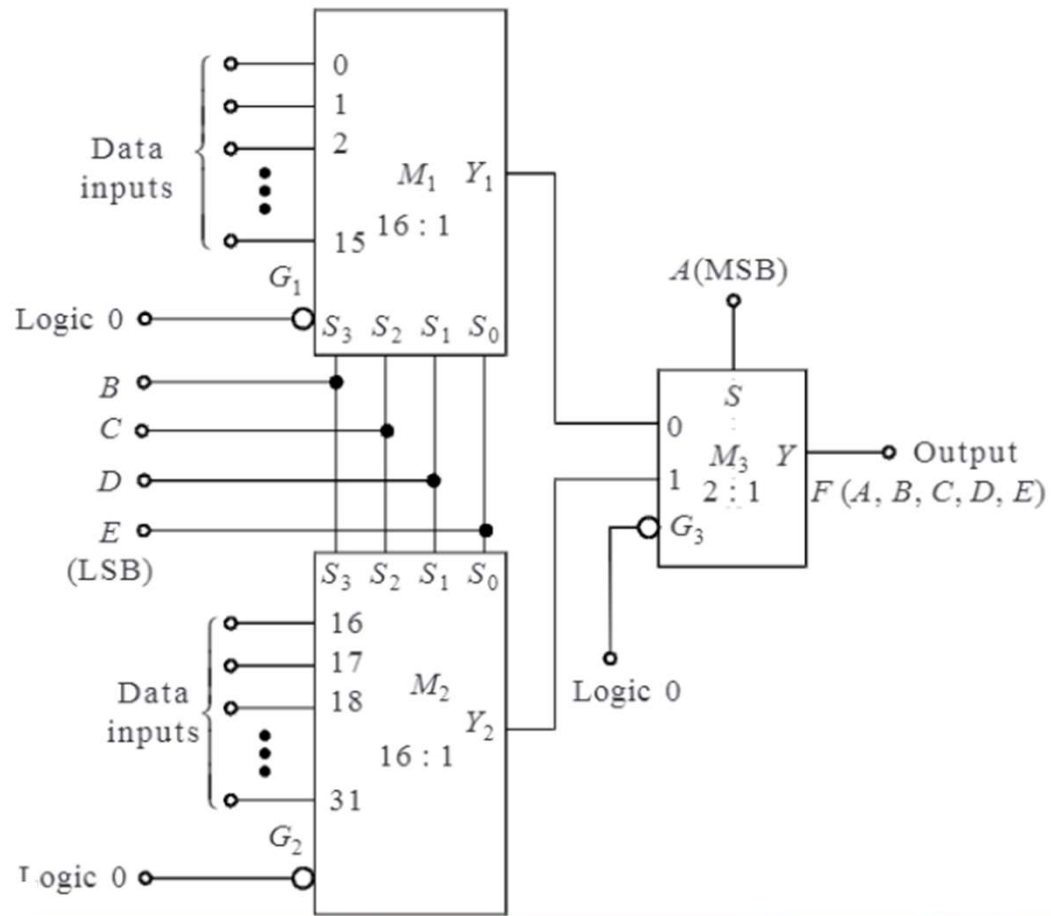
x y

- 0,3,4,5,11,12,14,15

Multiplexer Tree

- Can accommodate more inputs than available mux capacity.
- Standard available mux sizes- 2:1 mux (74157), 4:1mux (74153), 8:1mux (74151/ 74152) ,16:1mux (74150)
- A 32:1 mux can be created by cascading two 16:1 mux or
- five 8:1 mux lcs.





SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering



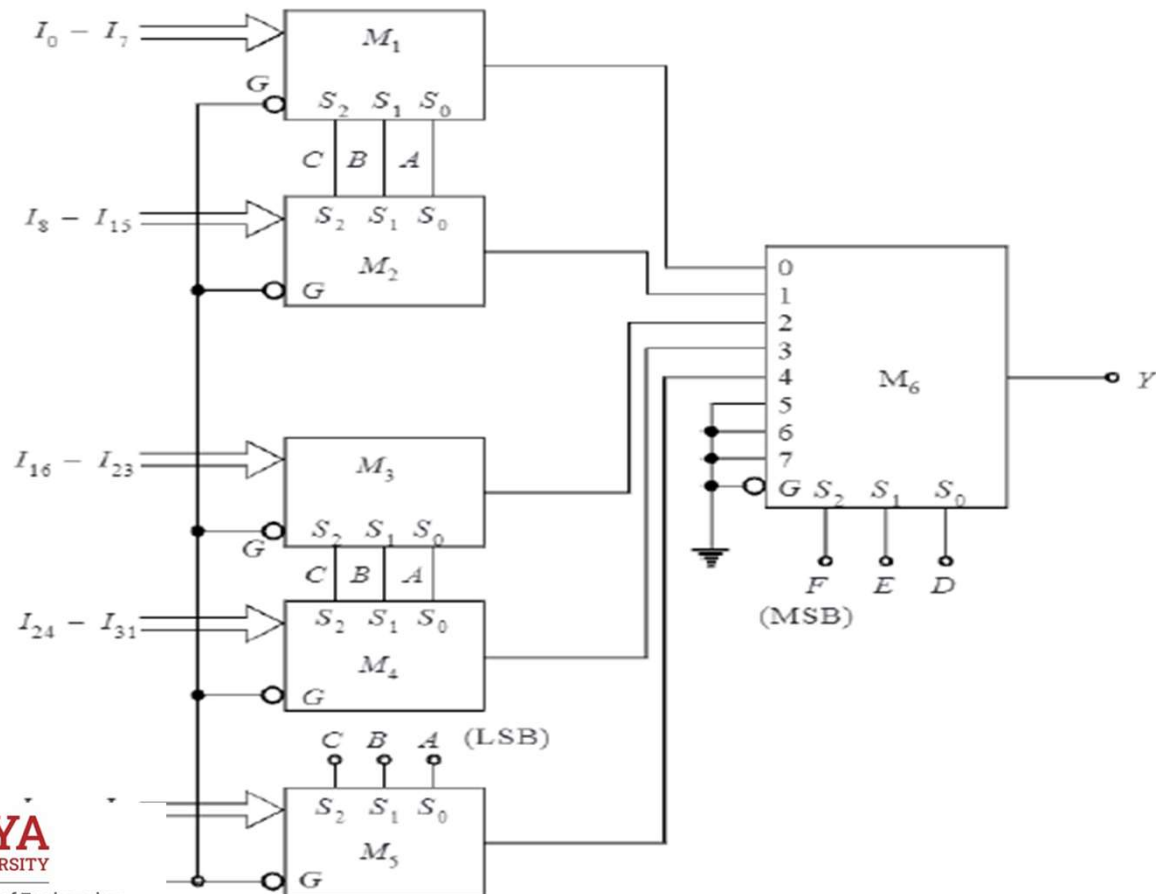


SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering



Design 40:1 mux using 8:1 mux





Demultiplexers

- One to many device
- Data Distributer
- 1:8 demux – 3 select lines

Decoders



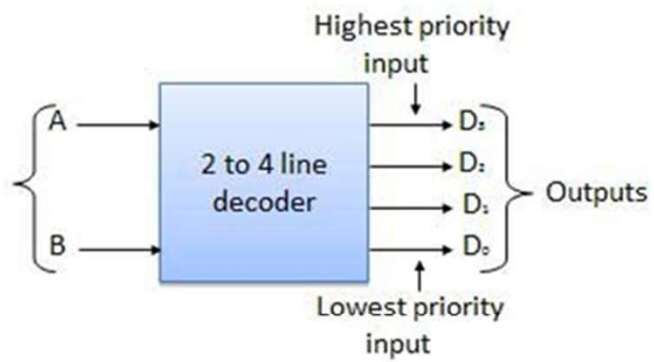
- 74139- dual 1:4 demux /decoder- inverted inputs
- 74139- 1:8 demux – Inverted input
- 74154- 1:16 demux



SOMAIYA
VIDYAVIHAR UNIVERSITY

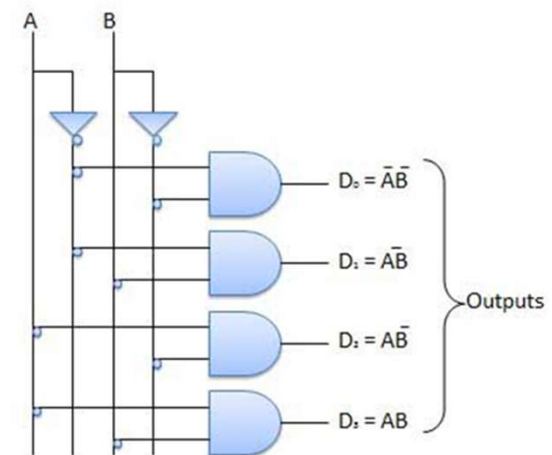
K J Somaiya College of Engineering





Truth Table

Inputs		Output			
A	B	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
0	1	0	0	1	0
1	1	0	0	0	1



Realization of logic function using decoder

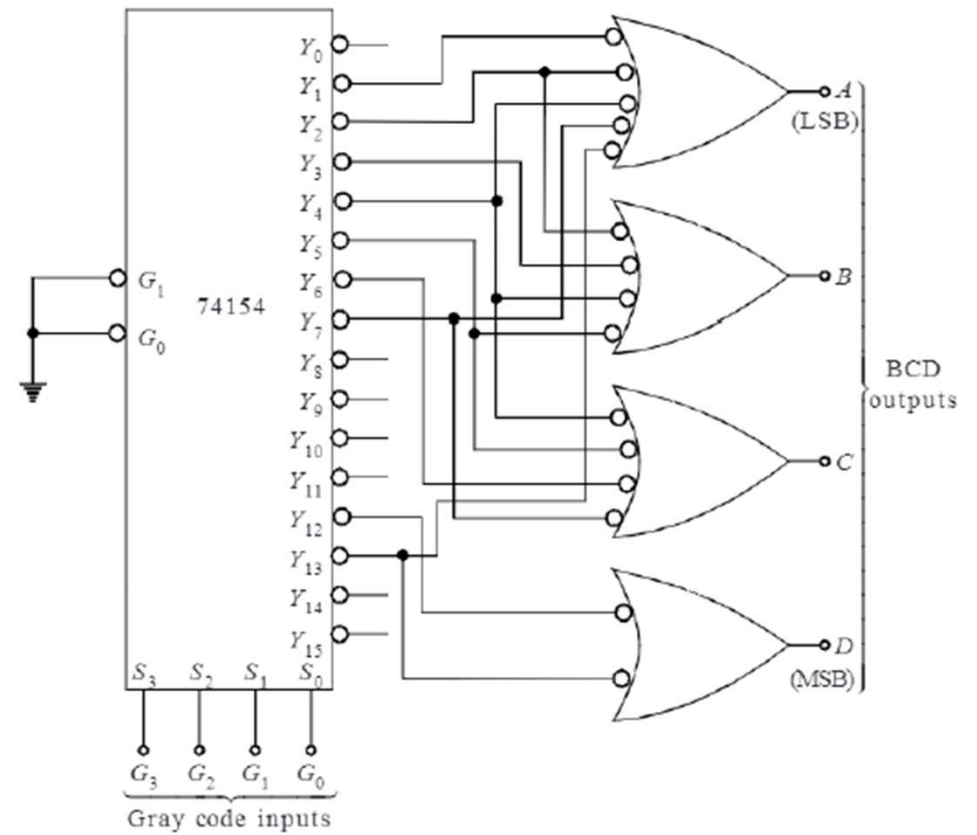
- $F_1 =$



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering

Gray code				BCD code			
G_3	G_2	G_1	G_0	D	C	B	A
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya College of Engineering